DUROTECH

DUROBENTONITE

Cold Applied, Multi-Layer, Hydroswelling Sheet Membrane

DuroBentonite is a uniquely reactive, hydrophilic waterproof membrane barrier. It comprises a robust multi-layer structure of hydro-swelling granular sodium bentonite locked between two layers of high strength geotextile fabric, one non-woven and the other woven reinforced. An HDPE barrier bonded to one surface is also available.

FEATURES

- Environmentally friendly - No hazardous ingredients
- Highly effective blind-side waterproofing barrier
- Immediate access for finishes
- Controlled uniform thickness
- Permanent waterproof barrier for life of structure
- Self-sealing to structural cracks and minor damage
- Seals fractures in land-fill sites
- Versatile and cost effective
- Bentonite gel laminates completely to substrate
- Simple, rapid installation
- Resistant to high hydrostatic pressures

USE AREA

DuroBentonite is specifically designed for critical blind-side tanking of structures in high water-table and subjected to intermittent or continuous hydrostatic pressure, including full boundary excavation using sheet piling, contiguous or secant piling, shotcrete and retention walls, under-slab. It is also suited to use in civil construction for driven tunnels as well as liquid containment in dams, ponds and land-fill. DuroBentonite may be installed to steel sheet pile, concrete, shotcrete, core-filled block and compacted substrates.

TECHNICAL INFORMATION

DuroBentonite Mass per Unit Area	4.8kg/m²
DuroBentonite Swelling Index	≥24 ml/g
Methylene Volume	32g/100g
Thickness	5mm
Roll Size	1.2m x 10m
Tensile Strength	>800N
Max Load Elongation	>10%
Peel Strength to Concrete	70N/100mm
Permeability Coefficient	3.9x10-11m/s
Hydrostatic Pressure	>70m
Puncture Resistance	>600N
Chemical, Gas and Bacterial Resistance	Excellent
Low Temp Flexibility @ -32°C	Unaffected

GRADES

- DuroBentonite GCL - DuroBentonite granules bound between two layers of tough non-woven geotextile. Use for all land-fill, containment ponds or general waterproofing.
- DuroBentonite ML - as GCL but with a tough surface layer of rain and damage resistant HDPE, a premium grade for major civil and commercial projects.

CHARACTERISTICS

- Active Compound - DuroBentonite is made of natural sodium-bentonite which is harmless to man and the environment - a truly Green Star waterproofing, having a VOC rating of zero.
- Waterproofing Mechanism - Sodium bentonite is nonmetallic montmorillonite clay formed from volcanic ash. It has a structure of layered flat platelets, ideally shaped to provide a hydraulic barrier. Sodium ions located between these platelets allows water to hydrate the bentonite to give its extraordinary swelling characteristics. It forms a high density, impermeable septum gel when confined in the presence of water.
  Used as a GCL, fully hydrated DuroBentonite achieves hydraulic conductivity of less than 5 x 10-9 cm/sec; about twenty times lower than a typical compacted clay layer
- Self-Healing - The self-healing properties of DuroBentonite are achieved by its reactive waterproof gel with a 20~28 times swelling volume. This enables DuroBentonite to seal cracks rapidly.
- Weather Resistance – Although recommended to be installed in a dry state, the unique fabrication confines any swelling from incidental contact by rain, within its composite structure.

PREPARATION:

Concrete blinding layer and other in-situ surfaces should be free of voids and sharp projections. Fill honeycomb concrete, she-bolt holes and other surface void with drypack mortar. Earth should be fully compacted. Substrates to be waterproofed must be dry and stable.

APPLICATION

GENERAL

Use 100 mm laps and stagger adjacent rolls at least 300 mm to avoid four-way laps. Laps should be 150 mm when shotcreting. Use nails and washers to fix in place and to give vertical support.

VERTICAL – BACKFILL (POST-APPLIED)

DuroBentonite is not recommended for Post Applied backfilled situations. In such cases use Grace Bituthene 3000 or Durotech Aquagard M. Contact Durotech Technical Department for specific design advice.

VERTICAL – BLIND SIDE (PRE-APPLIED)

Fix to formwork either horizontally or vertically in shingle fashion – overlap top over bottom. The woven fabric side should face installer. If using ML grade, HDPE side should be against substrate. Continue DuroBentonite to overlap 100 mm onto any previously applied DuroBentonite under-slab membrane. If DuroBentonite has not been used under-slab, apply a 40 mm fillet of DuroBentonite Mastic at wall/footing junction and continue membrane onto footing.

HORIZONTAL

Install DuroBentonite membrane to prepared substrate with woven geotextile side facing installer or the substrate to be waterproofed. Pin or staple laps as required, to prevent any displacement before and during concrete pour. Carry DuroBentonite through foundations, pile caps, ground beams, etc. Cut to fit around all penetrations and detail penetrations with 40 mm fillet of DuroBentonite Mastic. When connecting with sheet piling or secant/contiguous walls, return DuroBentonite installation a minimum 250 mm above finished floor slab or kicker level to allow the future connection of wall tanking. Cover this extension carefully withDurotech Protectoboard to prevent damage.

DETAILING AND REPAIR

Seal penetrations, anchors, ties or any damage with DuroBentonite Mastic or Powder. Use Durotech Protectoflash and DuroBentonite Mastic to seal perimeters and turn-ups.

NOTE

DuroBentonite must be kept in permanent compression. Does NOT use in unstable sub-strata conditions if compaction or blinding layer may later subside; or post applied to walls,as backfill may not maintain adequate compression.

COVERING

DuroBentonite should be covered as soon as possible after installation. It may be secured and confined in-place by any of the following systems:

POURED IN-PLACE CONCRETE

Place compact and finish as required on basement floors, retaining walls, lift-pits, tunnels, etc. Ensure DuroBentonite is not damaged or displaced during the placement process.

SHOTCRETE

Install and finish as required, care must be taken to ensure overlaps are not separated or damaged by concrete. The direction of spray should be across the lap shingle profile. Exercise care to prevent shadowing from reinforcing steel and creation of voids or boney areas.

PROTECTIVE SCREED

After placement of the DuroBentonite in horizontal situations, it may be optionally protected with a 50mm concrete screed.

LANDFILL & POND CONTAINMENT

Contact Durotech Technical Department for details.

COVERAGE & PACKAGING

- DuroBentonite – 1.2m x 10m x 5mm (12m^2) roll; up to 6m x 40m (240m) for civil projects. Allow 10 -15% waste.
- DuroBentonite Mastic – 20kg pail.
- DuroBentonite Powder – 20kg bag.
- DuroBentonite Waterstop 20mm – 10m rolls
- Chemind Protectoflash – 3m length

SHELF LIFE

Products will store indefinitely if kept in a cool dry place.

DUROTECH

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The information provided in this data sheet is correct at the time of printing (albeit is subject to change at any time) and is intended to give a simple description of the product and its capabilities. In practice, the substrate, intended surface to be treated and environmental conditions vary widely, making it essential for the user to determine the products suitability for a particular application and to ensure that the product is not used beyond its physical limitations. If in doubt contact the manufacturer. The product will perform as described herein provided it is applied in accordance with the manufacturer's instructions as stated in this data sheet and provided that the building and installation is structurally sound and the application is carried out competently. Durotech terms and conditions of sale apply.